Zhen Qin

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Biography. I am a 3nd-year Ph.D. Student at College of Computer Science and Technology, Zhejiang University, China, working on ***Federated Learning*** under the supervision of Prof. Shuiguang Deng. Currently, I mainly focus on 1) ***Federated Fine-tuning of Large Language Models***, 2) ***Personalized Federated Learning*** and 3) ***Trustworthy Federated Learning***. So far, I have published 9 papers on SIGKDD, AAAI, ICSOC, IEEE ICWS, IEEE TSC, KBS, etc.

Currently, I am working as a research intern at Damo Academy, Alibaba Group, focusing on ***Federated Fine-tuning of Large Language Models***. Prior to that, I worked as a research intern in 2012 Lab, Huawei Technologies Ltd. Co. from Mar. 2022 to May 2023, focusing on ***Preliminary Research of 6G architecture***.

Research Topics

Currently, I am focusing on several research topics in FL including :

- Federated LLM Fine-tuning
  Designing federated fine-tuning techniques for billion-sized large language models (LLMs), leveraging the vast quantities of data continuously generated at end devices to enhance the responsibility of LLMs to tasks described in natural language.
- Personalized Federated Learning
  Designing the personalized FL framework to provide adaptability to the statistical heterogeneity of data among distributed clients, to provide high model accuracy regardless of the degree of non-IIDness.
- Trustworthy Federated Learning
  - The defenses to backdoor attacks in FL through anomaly detection techniques.
  - Building fully decentralized FL systems on the basis of blockchain, providing distributed trustworthy for FL by extending existing Byzantine-tolerant FL approaches to decentralized scenarios and introducing extra privacy-preserving technologies.

Education

Sep 2021 – Mar 2025 (expected)	PhD Student (*Computer Science and Technology*) College of Computer Science and Technology, Zhejiang University (ZJU), Hangzhou, China Supervisor : Prof. Shuiguang Deng
Sep 2018 – Jun 2021	Master of Science in Engineering (*Computer Application Technology*) School of Computer Engineering and Science, Shanghai University (SHU), Shanghai, China Supervisor : Prof. Guobing Zou Score : 90.0/100
Sep 2014 – Jun 2018	Bachelor of Engineering (*Computer Science and Technology*) School of Computer Engineering and Science, Shanghai University (SHU), Shanghai, China GPA : 3.22/4.0 | Rank : Top 20% Minor in Economics at East China Normal University (ECNU), Shanghai, China

Publications (First Author or First Student Author)

- Zhen Qin, Daoyuan Chen, Bingchen Qian, Bolin Ding, Yaliang Li*, Shuiguang Deng*. Federated Full-Parameter Tuning of Billion-Sized Language Models with Communication Cost under 18 Kilobytes. arXiv:2312.06353, 2023. **[PDF] [Code]**
- Zhen Qin, Feiyi Chen, Chen Zhi, Xueqiang Yan, Shuiguang Deng*. Resisting Backdoor Attacks in Federated Learning via Bidirectional Elections and Individual Perspective. accepted by Annual AAAI Conference on Artificial Intelligence (AAAI), 2023. **(CCF A)** **[PDF][Code]**
- Zhen Qin, Shuiguang Deng*, Mingyu Zhao, Xueqiang Yan. FedAPEN : Personalized Cross-silo Federated Learning with Adaptability to Statistical Heterogeneity. ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD), pp. 1954-1964, 2023. **(CCF A)** **[PDF][Code]**
- Zhen Qin, Shuiguang Deng*, Xueqiang Yan, Lu Lu, Mingyu Zhao, Yan Xi, Jianjun Wu, Tao Sun, Nanxiang Shi. 6G Data Plane : A Novel Architecture Enabling Data Collaboration with Arbitrary Topology. Mobile Networks and

Applications, vol.28, pp. 394–405, 2023. **(CCF C, IF=3.077) [PDF]**

- Zhen Qin, Xueqiang Yan, Mengchu Zhou, Shuiguang Deng*. BlockDFL : A Blockchain-based Fully Decentralized Federated Learning Framework. arXiv:2205.10568, 2023. **(Under Review) [PDF]**
- Guobing Zou, Zhen Qin, Shuiguang Deng, Kuan-Ching Li, Yanglan Gan*, Bofeng Zhang*. Towards the Optimality of Service Instance Selection in Mobile Edge Computing. Knowledge-Based Systems, vol. 217, p. 106831, 2021. **(CCF C, IF=8.139) [PDF]**
- Guobing Zou, Zhen Qin, Qiang He, Pengwei Wang, Bofeng Zhang, Yanglan Gan*. DeepWSC : Clustering Web Services via Integrating Service Composability into Deep Semantic Features. IEEE Transactions on Services Computing, vol. 15, no. 4, pp. 1940-1953, 2021. **(CCF A, IF=11.019) [PDF]**
- Guobing Zou, Zhen Qin, Qiang He, Pengwei Wang, Bofeng Zhang, Yanglan Gan*. DeepWSC : A Novel Framework with Deep Neural Network for Web Service Clustering. IEEE International Conference on Web Services (ICWS), 2019, pp. 434-436. **(CCF B) [PDF]**

Publications (Co-authored)

- Feiyi Chen, Yingying Zhang, Zhen Qin, Lunting Fan, Renhe Jiang, Yuxuan Liang, Qingsong Wen, Shuiguang Deng*. MACE : A Multi-pattern Accommodated and Efficient Anomaly Detection Method in the Frequency Domain. arXiv:2311.16191, 2023. **(Under Review) [PDF]**
- Feiyi Chen, Zhen Qin, Yingying Zhang, Shuiguang Deng*, Yi Xiao, Guansong Pang, Qingsong Wen. LARA : A Light and Anti-overfitting Retraining Approach for Unsupervised Anomaly Detection. arXiv:2310.05668, 2023. **(Under Review) [PDF]**
- Feiyi Chen, Zhen Qin, Hailiang Zhao, Mengchu Zhou, Shuiguang Deng*. PePNet : A Periodicity-Perceived Workload Prediction Network Supporting Rare Occurrence of Heavy Workload. arXiv:2308.01917, 2023. **(Under Review) [PDF]**
- Guobing Zou, Ya Liu, Zhen Qin, Jin Chen, Zhiwei Xu, Yanglan Gan*, Bofeng Zhang, Qiang He*. ST-EUA : Spatio-temporal Edge User Allocation with Task Decomposition. IEEE Transactions on Services Computing, vol. 16, no. 1, pp. 628-641, 2022. **(CCF A, IF=11.019) [PDF]**
- Guobing Zou, Ya Liu, Zhen Qin*, Jin Chen, Zhiwei Xu, Yanglan Gan, Bofeng Zhang, Qiang He*. TD-EUA : Task-decomposable Edge User Allocation with QoE Optimization. International Conference on Service Oriented Computing (ICSOC), 2020, pp. 215-231. **(CCF B) [PDF]**

Internship Experience

Jun 2023 – Now Damo Academy, Alibaba Group (*Research Intern*)

- Federated Fine-tuning of LLMs : Exploring the possibilities of tuning large models based on federated learning, mainly addressing communication overhead and memory cost issues.
- System Development : Exploring memory-efficient fine-tuning techniques for LLMs which are suitable for cross-device FL, and implementing them based on FederatedScope.

`Federated Learning` `Large Language Model` `Development`

Mar 2022 – May 2023 2012 Lab, Huawei Technologies, Ltd. Co. (*Research Intern*)

- Network Architecture : Investigate existing network architectures; and design 6G network architecture according to the visions of 6G network architecture.
- Collaboration Architecture : Designing edge-edge collaboration frameworks for decentralized FL.
- White Paper : Participating in the writing of two white papers for 6G ANA, i.e., i), involving with *6G Data Service* and ii) *Wireless Network Data Dictionary*, respectively.

`Federated Learning` `6G Architecture` `Blockchain`

Honors & Awards

Nov 2023	Outstanding Graduate Student of Zhejiang University in 2023
Nov 2022	Outstanding Graduate Student of Zhejiang University in 2022
Mar 2021	Excellent Graduate of Shanghai
Oct 2020	National Scholarship for Graduate Students (in Shanghai University)
Dec 2019	National Scholarship for Graduate Students (in Shanghai University)
Dec 2018	Second Prize of China Post-graduate Mathematical Contest in Modeling (<14.5%)

(Last Update : 18 Dec. 2023)